Statistical Learning and Data Analysis Module B

Logistic Regression

Roberta Siciliano

Statistics, tèchnè-loghìa, analysis of data

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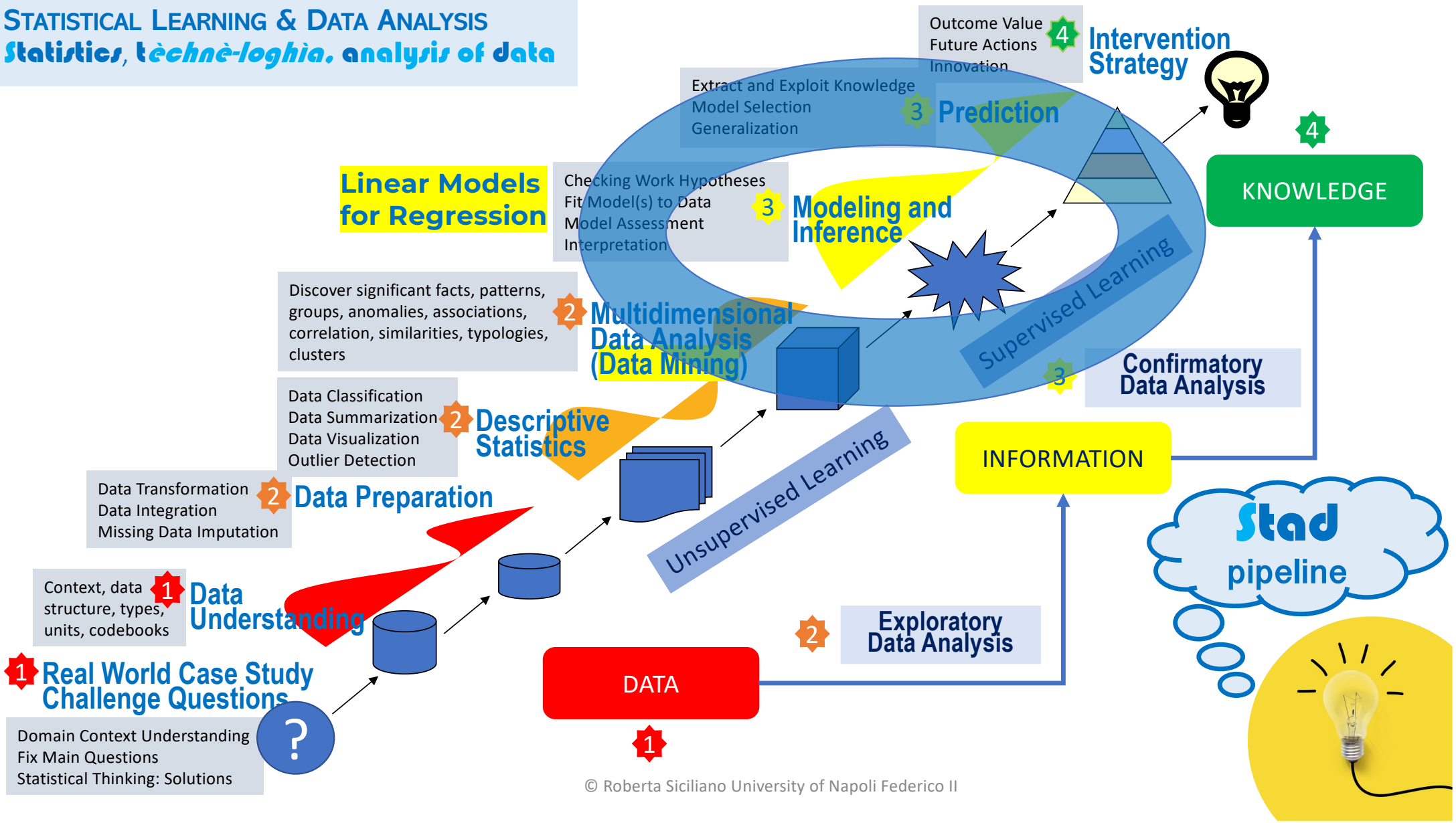
Statistical Learning & Data Analysis

Statistics, tèchnè-loghìa, analysis of data



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A thought

«Since childhood we classify objects and learn what are the facts that discriminate different groups of objects.»

«It is supervised learning when a parent shows the child the class to which each object belongs so that he can learn»

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An Overview on Classification

- Classifying means assigning an observation to a category or class.
- In Supervised Classification there is a target Y for this assignment and this target is qualitative or categorical.
- Classification problems occur often and even more than regression problems.
- Very often Y is binary assuming values $0/1$.

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Classifiers

Most widely classification techniques:

- **Logistic Regression**
- *Generative Models (Discriminant Analysis)*
- *K-Nearest Neighbors*

More computer-intensive methods:

- *Generalized Additive Models*
- *Decision Trees*
- *Random Forests (Ensemble method)*
- *Boosting (Ensemble method)*
- *Support Vector Machines*
- Artificial Neural Networks

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The problem

A person arrives at the emergency room with a set of symptoms that could possibly be attributed to one of four medical conditions. *Which of the four conditions* does the individual have?

An online banking service must be able to determine *whether or not* a transaction being performed on the site is fraudulent, on the basis of the user's IP address, past transaction history, and so forth.

On the basis of DNA sequence data for a number of patients *with and without* a given disease, a biologist would like to figure out which DNA mutations are deleterious (disease-causing) and which are not.

The transportation system planning depends on the travel demand. Which are the most discriminant characteristics about the individuals to prefer the public travel model with respect to the private one? Or to determine *the preference among more alternatives* such as Walking, Bicycle, Public Transport, Car?

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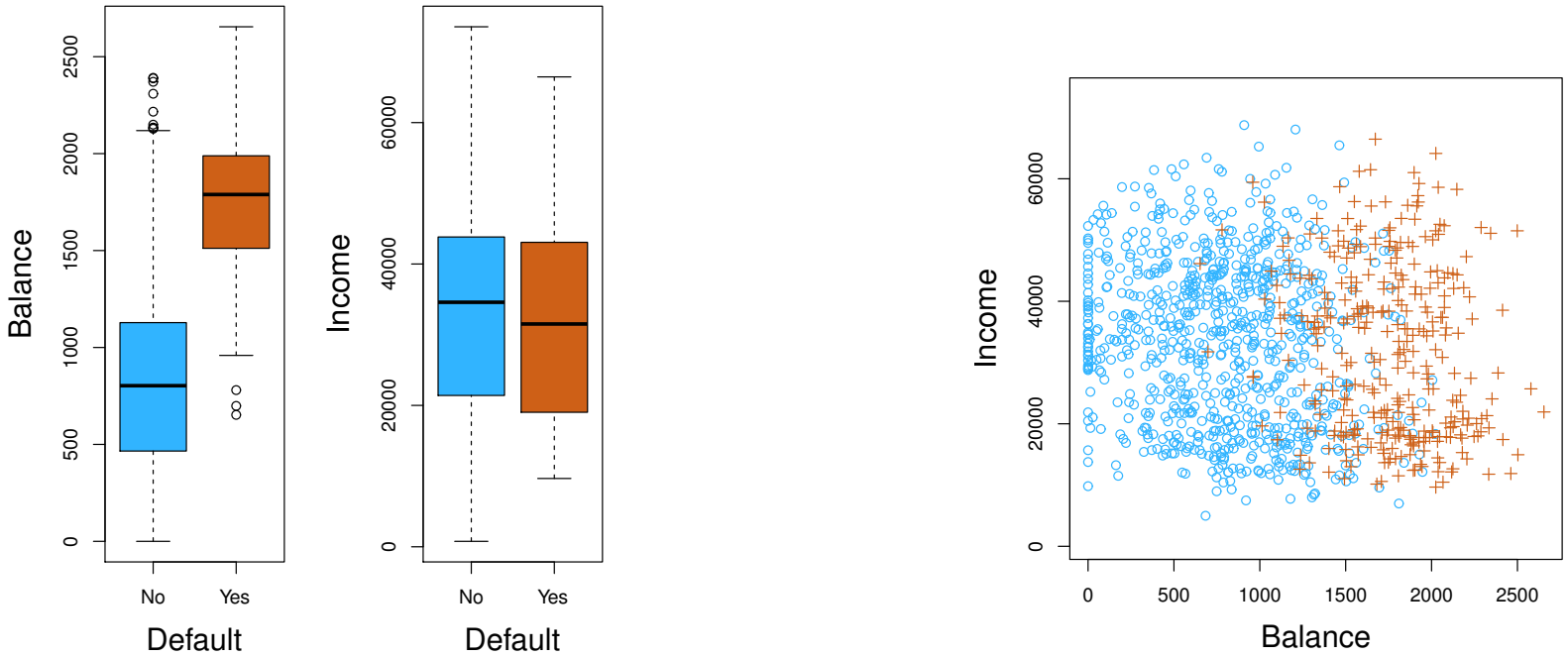

Binary Target

Default data set:

- Default (YES/NO) *versus* Annual Income and Monthly Credit Card Balances of a number of individuals

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Logistic Regression

- Why Not Linear Regression
- Logistic Regression Model
- Estimating the Regression Coefficients
- Making Predictions
- Multiple Logistic Regression
- Multinomial Classification

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Why Not Linear Regression?

- Linear Regression is not appropriate in the case of qualitative Y.
- For a binary target Y the error component can be a Bernoulli random variable and thus also Y is a Bernoulli random variable with values $Y = 0$ (unsuccess) or $Y = 1$ (success).

$$Y = E(Y|X = x) + \varepsilon$$

- The systematic component is the probability of success that in linear regression should be "forced" to be linear

$$E(Y|X = x) = p(X) = \beta_0 + \beta_1 X$$

- Fitting OLS the estimates $\hat{y}_i$ do not necessarily fall into the probability range $[0,1]$.
- Thus, $E(Y|X = x) = p(X)$ must follow a probability distribution function such as:
  - Logistic → LOGIT (Logistic Regression)
  - Normal → PROBIT

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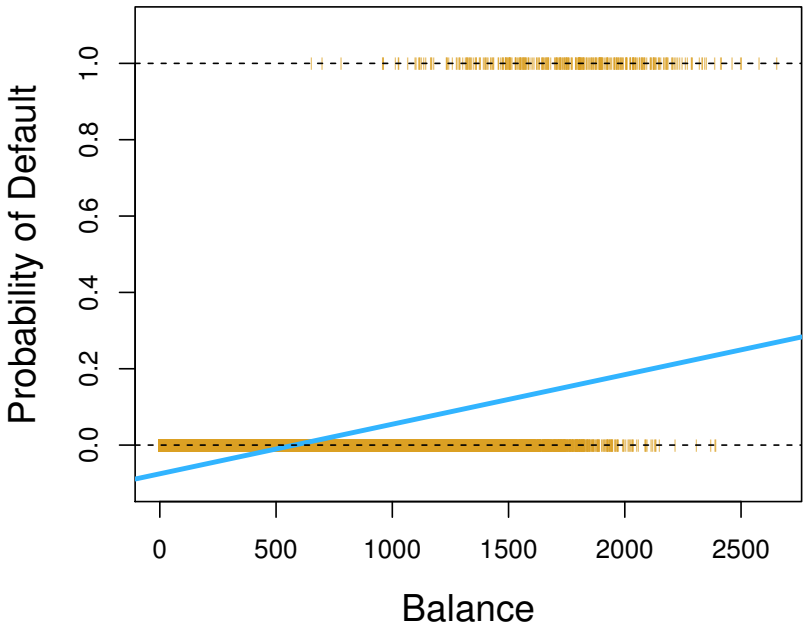
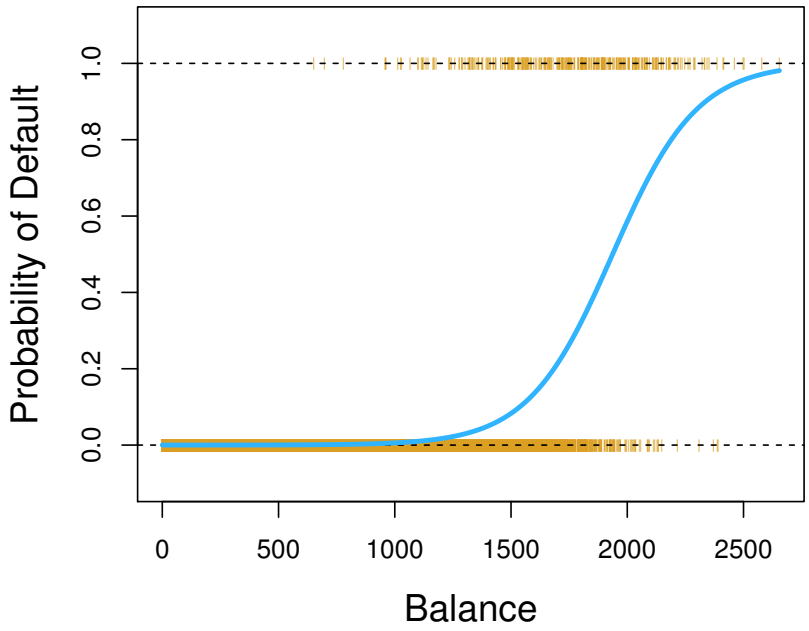
Linear versus Logistic Regression





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Multi-Class Target

- Try to predict the medical condition of a patient at the emergency room on the basis of him/her symptoms.
- Suppose to code four possible diagnoses (response classes) with values:
  - $Y = 1$ `if stroke,`
  - $Y = 2$ `if drug overdose,`
  - $Y = 3$ `if epileptic seizure,`
  - $Y = 4$ `if other.`
- This means to fix an ordering on the outcomes and some constraints on the OLS solutions, say the difference between `stroke` and `drug overdose` is the same as the difference between `drug overdose` and `epileptic seizure` and so on.

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Logistic Regression Model

- The logistic function

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}$$

- Odds

$$\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X} \geq 0$$

✓*Example: On average* **1 in 5 people** *will default means that* $p(X) = 0.2$ *and* $\frac{0.2}{1-0.2} = \frac{1}{4}$.

- Log-odds or Logit

$$logit\big(p(X)\big) = log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X$$

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Interpretation of parameters

- The **regression coefficients**

✓ $logit\big(p(X = x + 1)\big) - logit\big(p(X = x)\big) = \beta_0 + \beta_1(x + 1) - (\beta_0 + \beta_1 x) = \beta_1$

Increasing X by one unit changes the log odds by β_1 or, equivalently, it multiplies the odds by e^{β_1} .

✓ β_0 is the odds when the predictors have no effect.

- The **Odds ratio**

$$\frac{p_1(X)/\big(1 - p_1(X)\big)}{p_2(X)/\big(1 - p_2(X)\big)}$$

- If the odds ratio is equal to 1, the odds $\frac{p(X)}{1-p(X)}$ is the same for the individual 1 and for the individual 2. In the other cases, the probability $p(X)$ differs from one individual to another.

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Estimating the Regression Coefficients

- Maximizing the Likelihood function

$$L(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=o} \left(1 - p(x_{i'})\right)$$

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} p(x_i)^{y_i} \left(1 - p(x_i)\right)^{1-y_i}$$

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^{n} \left\{ y_i log\left(p(x_i)\right) + (1 - y_i) log\left(1 - p(x_i)\right) \right\}$$

- Iterative methods such as Newton-Raphson.

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To verify the significance of a single coefficient

- The Wald statistic per the j-th regression coefficient:

$W = \frac{\hat{\beta}_j - \beta_j}{s.e.(\hat{\beta}_j)} \sim N(0,1)$ for large samples

- The output reports:

✓ The estimated regression coefficient $\hat{\beta}_j$ with its standard error $s.e.(\hat{\beta}_j)$

✓ The empirical test value z_{emp} to verify $H_0: \beta_j = 0$ with the associated p -value $P(|Z| \geq |z_{emp}|) = p$

➢ For $p \leq \alpha$ the null hypothesis is to be reject and the predictor is significant for the model

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Accuracy of the estimates

Default data set:

- Default (YES/NO) *versus* Monthly Credit Card Balances of a number of individuals

	Coefficient	**Std. error**	**Z-statistic**	**P-value**
Intercept	-10.6513	0.3612	-29.5	<0.0001
balance	0.0055	0.0002	24.9	<0.0001

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Accuracy of the estimates

Default data set:

- Default (YES/NO) *versus* Student status (dummy variable) of a number of individuals

	Coefficient	Std. error	Z-statistic	P-value
Intercept	-3.5041	0.0707	-49.55	<0.0001
Student[Yes]	0.4049	0.1150	3.52	0.0004

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Making Predictions

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 1{,}000}}{1+e^{-10.6513+0.0055\times 1{,}000}} = 0.00576$$

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 2{,}000}}{1+e^{-10.6513+0.0055\times 2{,}000}} = 0.586$$

$$\hat{p}(\text{default} = \text{Yes}|\text{student} = \text{Yes}) = \frac{e^{-3.5041+0.4049\times 1}}{1+e^{-3.5041+0.4049\times 1}} = 0.0431$$

$$\hat{p}(\text{default} = \text{Yes}|\text{student} = \text{No}) = \frac{e^{-3.5041+0.4049\times 0}}{1+e^{-3.5041+0.4049\times 0}} = 0.0292$$

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Challenger Space Shuttle - O-Ring Data

UCI Machine Learning Repository (http://archive.ics.uci.edu/ml/)

- In 1986 a Space Shuttle exploded during the take off phase. Later it was ascertained that one of the possible causes of the disaster was the failure suffered by a gasket (O-Ring) caused by the operating temperature of the aircraft.
- The temperatures in degrees Fahrenheit were recorded at the take-off of all the previous 23 shuttle flights and the anomalies suffered by the seals.
- The dataset is made up of the following variables:
  - Temperatures -> take-off temperature expressed in degrees Fahrenheit;
  - Damage -> Damage suffered by the seals (0 = no damage, 1 = damage)

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The problem and the expected solution

- Through logistic regression model it is possible to investigate the effect of the temperature of the aircraft at take-off on the probability that a gasket could be damaged

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```
Iterazione 0: log verosimiglianza = -16,2304090747
Iterazione 1: log verosimiglianza = -10,5549358983
Iterazione 2: log verosimiglianza = -10,1759403489
Iterazione 3: log verosimiglianza = -10,1576738930
Iterazione 4: log verosimiglianza = -10,1575963457
Iterazione 5: log verosimiglianza = -10,1575963439

Convergenza raggiunta dopo 6 iterazioni

Modello 2: Stime Logit usando le 23 osservazioni 1-23
Variabile dipendente: Damaged

      VARIABILE       COEFFICIENTE       ERRORE STD    STAT T    P-VALUE

  const                 15,0429            7,37864       2,039   0,04148 **
  Temperature           -0,232163          0,108237     -2,145   0,03196 **

  Media di Damaged = 0,304
  Numero dei casi 'previsti correttamente' = 20 (87,0%)
  Pseudo-R-quadro di McFadden = 0,281315
  Log-verosimiglianza = -10,1576
  Test del rapporto di verosimiglianza: Chi-quadro(1) = 7,95196 (p-value 0,004804)
  Criterio di informazione di Akaike (AIC) = 24,3152
  Criterio bayesiano di Schwarz (BIC) = 26,5862
  Criterio di Hannan-Quinn (HQC) = 24,8863

               Previsto
                 0    1
  Effettivo 0   16    0
            1    3    4
```

How to interpret the solution?

The probability that a seal may be damaged:

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{15.0429-0.2332\cdot X}}{1+e^{15.0429-0.2332\cdot X}}$$

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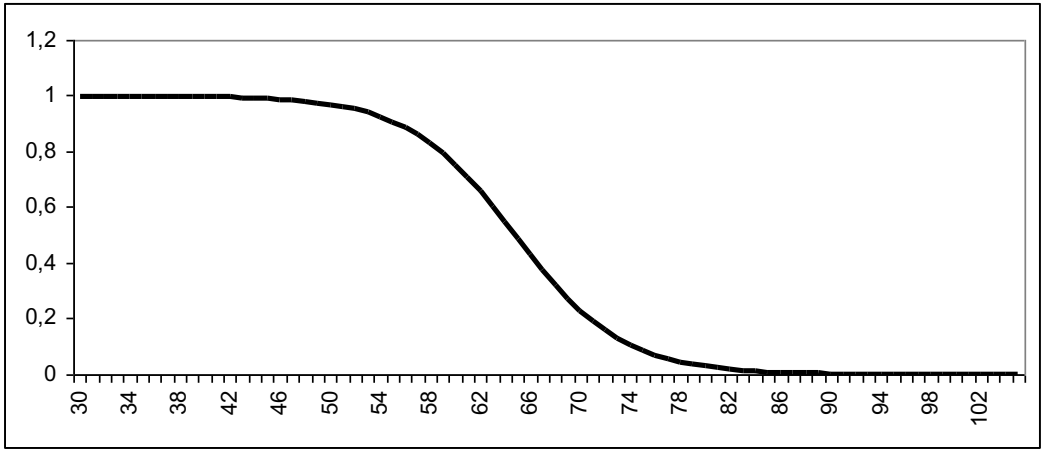
The beta coefficient is negative.



The probability of success (the seal is broken) decreases as the values of X (temperature) increase.

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Making Predictions

- What is the probability of the seal breaking at a temperature of 31 degrees Fahrenheit?
- $\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{15.0429-0.2332\cdot 31}}{1+e^{5.0429-0.2332\cdot 31}} = 0.9996$

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Multiple Logistic Regression

- $log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$
- $p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0++\beta_1 X_1+\cdots+\beta_p X_p}}$

- Predictors can be numerical and/or categorical (using binary variables as in linear regression)
- Model can be specified with or without interactions (as in linear regression)

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Fitting by Maximum Likelihood Method

- Maximizing the Log-Likelihood function

$$\ell(\beta) = \sum_{i=1}^{n} \{y_i log\big(p(x_i;\beta)\big) + (1-y_i) log\big(1-p(x_i;\beta)\big)\}$$

$$\ell(\beta) = \sum_{i=1}^{n} \left\{y_i \beta^T x_i + log\left(1+e^{\beta^T x_i}\right)\right\}$$

where β includes the intercept and x_i includes the constant term 1.

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The score equations

- The *score* equations are:

$$\frac{\partial \ell(\beta)}{\partial \beta} = \sum_{i=1}^{n} x_i \big(y_i - p(x_i;\beta)\big) = 0$$

which are $p+1$ equations nonlinear in β.

- Since the first term in x_i is 1, the first equation specifies that $\sum_{i=1}^{n} y_i = \sum_{i=1}^{n} p(x_i;\beta)$, the expected number of class ones matches the observed number (and hence also class 0).

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Newton-Raphson Algorithm

- It is an iterative numerical algorithm which requires the second derivative or Hessian matrix

$$\frac{\partial^2 \ell(\beta)}{\partial\beta\partial\beta^T} = -\sum_{i=1}^{n} x_i x_i^T p(x_i;\beta)\big(1 - p(x_i;\beta)\big)$$

- Starting with β^{old}, a single Netwton-Raphson update is:

$$\beta^{new} = \beta^{old} - \left(\frac{\partial^2 \ell(\beta)}{\partial\beta\partial\beta^T}\right)^{-1} \frac{\partial \ell(\beta)}{\partial\beta}$$

where the derivatives are evaluated at β^{old}.

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Some matrix notation

- $\mathbf{y}$: the vector of y_i values
- $\mathbf{X}$: the $n \times (p+1)$ matrix of x_i values
- $\mathbf{p}$: be the vector of fitted probabilities with the i-th element $p\left(x_i; \beta^{old}\right)$
- $\mathbf{W}$: the diagonal matrix with the i-th element $p\left(x_i; \beta^{old}\right)\left(1 - p\left(x_i; \beta^{old}\right)\right)$
- $\frac{\partial \ell(\beta)}{\partial \beta} = \mathbf{X}^T(\mathbf{y} - \mathbf{p})$: the score equations
- $\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T} = -\mathbf{X}^T\mathbf{W}\mathbf{X}$: the Hessian

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Iteratively Reweighted Least-Squares (IRLS)

The Newton-Raphson step is thus

$$\beta^{new} = \beta^{old} + (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{y}-\mathbf{p}) =$$

$$= (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\left(\mathbf{X}\beta^{old} + \mathbf{W}^{-1}(\mathbf{y}-\mathbf{p})\right) =$$

$$= (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\,\mathbf{z}$$

which can be understood as a *weighted least squares step* with the response

$$\mathbf{z} = \mathbf{X}\beta^{old} + \mathbf{W}^{-1}(\mathbf{y}-\mathbf{p})$$

called *Adjusted response.*

This algorithm is referred to as **Iteratively Reweighted Least Squares (IRSL)** since each iteration solves the weighted least squares problem:

$$\beta^{new} \leftarrow \underset{\beta}{\operatorname{argmin}}(\mathbf{z}-\mathbf{X}\beta)^T\mathbf{W}(\mathbf{z}-\mathbf{X}\beta)$$

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	Coefficient	Std. error	Z-statistic	P-value
Intercept	-10.8690	0.4923	-22.08	<0.0001
Balance	0.0057	0.0002	24.74	<0.0001
Income	0.0030	0.0082	0.37	0.7115
Student[Yes]	-0.6468	0.2362	-2.74	0.0062

Accuracy of the estimates in multiple logistic regression model

Default data set:

- Default (YES/NO) *versus* Annual Income, Monthly Credit cards Balance, Student status (dummy variable) of a number of individuals

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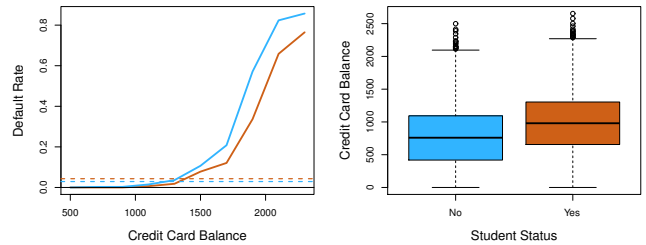
Confounding effect

- The negative coefficient for student in the multiple logistic regression indicates that *for a fixed value of balance and income*, a student is less likely to default than a non-student.
- **The variables student and balance are correlated.**
- Students tend to hold higher levels of debt, which is in turn associated with higher probability of default. In other words, students are more likely to have large credit card balances.
- A student is riskier than a non-student if no information about the student's credit card balance is available. However*, that student is less risky than a non-student with the same credit card balance!*



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Making Predictions

$$\hat{p}(X) = \frac{e^{-10.869+0.00574\times 1{,}500+0.003\times 40-0.6468\times 1}}{1+e^{-10.869+0.00574\times 1{,}500+0.003\times 40-0.6468\times 1}} = 0.058$$

$$\hat{p}(X) = \frac{e^{-10.869+0.00574\times 1{,}500+0.003\times 40-0.6468\times 0}}{1+e^{-10.869+0.00574\times 1{,}500+0.003\times 40-0.6468\times 0}} = 0.105$$

The income coefficient estimate is multiplied by 40 rather than 40,000 because in the model fit the estimates are in units of $1,000.

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A look at the Goodness of Fit

- The goodness of fit of a model to data can be interpreted as a way to replace a set of observed values with a set of estimated values obtained from a model that uses a small number of parameters.
- The model estimates are required to be as close as possible to the observed values of the response variable, that is, the discrepancy between these values must be minimal.
- Inferential procedures aim
  - *To verify the goodness of the model as a whole.*
  - *To verify the goodness of a "parsimonious" model*
  - *To verify the difference between two comparable models.*

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To verify the goodness of the model as a whole

- McCullagh-Nelder Deviation based on the Maximum Likelihood Ratio

$$D = -2log\left[\frac{\max\limits_{\hat{\beta}} L(M)}{\max\limits_{\hat{\beta}} L(M_{sat})}\right] \sim \chi^2_{(n-p-1)}$$

- where M is the fitted model and M_{sat} is the saturated model.
- If the fitted model M is good, then its likelihood will be very close to that of the saturated model M_{sat} which represents its maximum value.
- Consequently, small values of D imply similar values of their likelihoods and therefore a good adaptation of the model.

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To verify the goodness of a "parsimonious" model

- A model is good when it expresses the balance between the best possible fit and the least complexity of the fitted model.
- The D statistic considers only the best possible fit which is not necessarily a parsimonious model.
- The G statistic looks for a parsimonious model:

$$G = -2log\left[\frac{\max_{\hat{\beta}} L(M_0)}{\max_{\hat{\beta}} L(M)}\right] \sim \chi^2_{(p)}$$

where M is the fitted model and M_0 is the trivial model with only the intercept.

- It holds: $G = D(M_0) - D(M)$

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To verify the difference between two comparable models

- It is possible to compare two models, where one includes a subset of predictors, say $(p-q)$.
- The G statistic (called extra-deviation) is

$$G = -2log\left[\frac{\max_{\hat{\beta}} L\left(M_{p-q}\right)}{\max_{\hat{\beta}} L(M)}\right] \sim \chi^2_{(q)}$$

- If the extra-deviance is significantly large then the choice will fall on the model with p predictors, vice versa if G is small then it is better to opt for the reduced (and more parsimonious) model.

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Multinomial Logistic Regression

- **Single class as baseline (the K-th):** estimate $K-1$ sets of parameters

$$P(Y=k|X=x)=\frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{1+\sum_{l=1}^{K-1}e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}} \quad \text{for } k=1,\dots,K-1$$

$$P(Y=K|X=x)=\frac{1}{1+\sum_{l=1}^{K-1}e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

so that

$$log\left(\frac{P(Y=k|X=x)}{P(Y=K|X=x)}\right)=\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p$$

- **Softmax coding:** estimate K sets of parameters

$$P(Y=k|X=x)=\frac{e^{\beta_{k0}+\beta_{k1}x_1+\cdots+\beta_{kp}x_p}}{\sum_{l=1}^{K}e^{\beta_{l0}+\beta_{l1}x_1+\cdots+\beta_{lp}x_p}}$$

so that

$$log\left(\frac{P(Y=k|X=x)}{P(Y=k'|X=x)}\right)=(\beta_{k0}-\beta_{k0})+(\beta_{k1}-\beta_{k1})x_1+\cdots+(\beta_{kp}-\beta_{kp})x_p$$

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Multinomial Classification

- Discrete Choice Models
- Random Utility Theory (Mc Fadden)
- Independence of Irrelevant Alternatives
- Multinomial Logit Model
- Nested Logit Model
- Cross-Nested Logit Model
- Latent Class Model
- Mixed Logit Model

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